Blinkit Logistics Optimization & Supply Chain Analytics

Quick Commerce (Retail & Logistics) • Team ID: E_G-16

Faculty Mentor: Ms. Aayushi Vashishth

- GitHub Repository URL: [Link]
- Tableau Public URL: [Link]

Team Members:
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Context & Problem Statement

Sector Context:

- In Quick Commerce, logistical efficiency drives unit economics.
- Delays do not just cause complaints; they cause permanent customer churn.

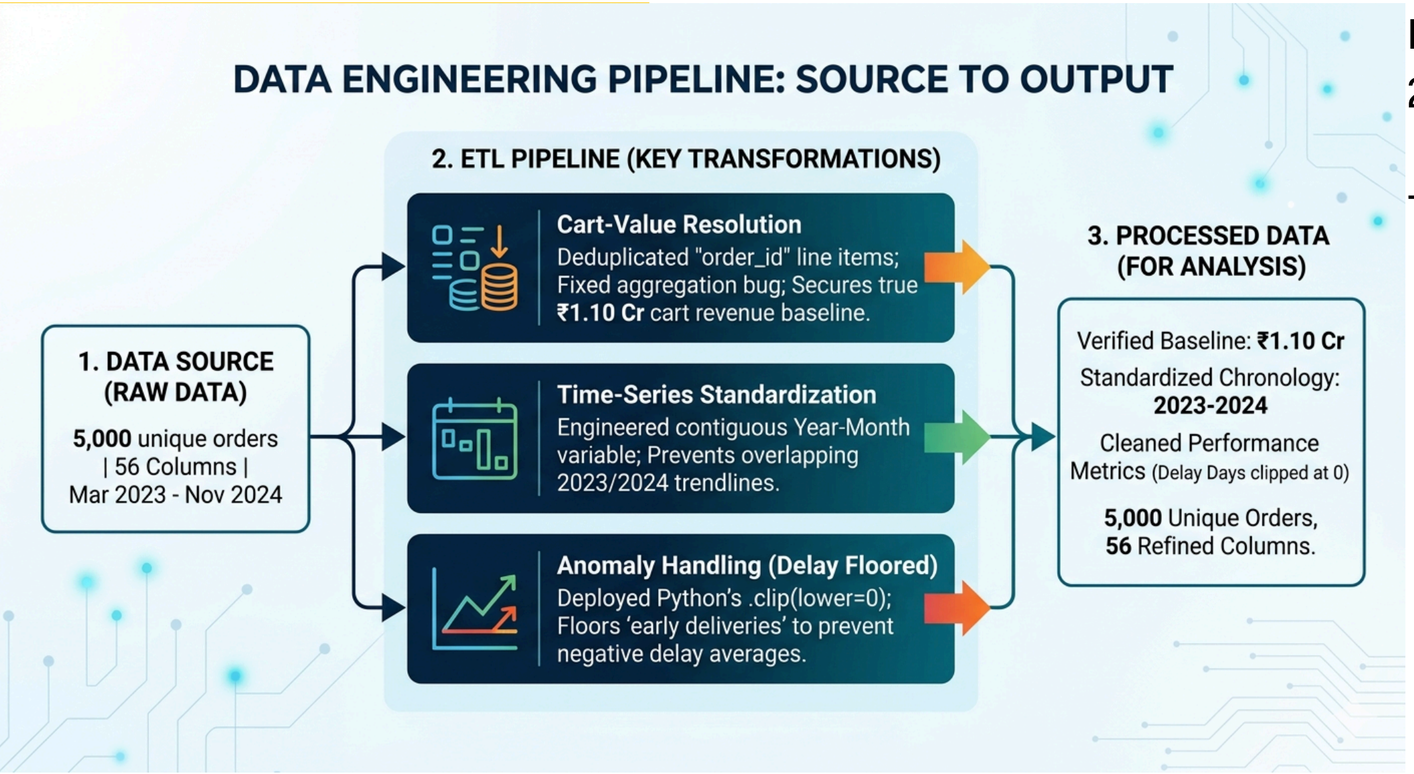
Problem Statement:

How do delivery bottlenecks impact high-value customer retention, and how can rider allocation and inventory management be optimized to protect the bottom line?

Objective:

Identify the operational root causes of delays and deploy data-driven workforce and inventory strategies.

Data Engineering (Source to Pipeline)



Data Source: 5,000 unique orders | 56 Columns | Mar 2023 - Nov 2024.

The ETL Pipeline (Key Transformations):

- Cart-Value Resolution: Deduplicated order_id line items to fix an aggregation bug, securing the true ₹1.10 Cr cart revenue baseline.
- Time-Series Standardization: Engineered a contiguous Year-Month variable to fix overlapping 2023/2024 trendlines.
- Anomaly Handling: Deployed Python's .clip(lower=0) to floor early deliveries, preventing negative values from skewing the delay average.

KPI & Metrics Framework

Core Measured Metrics:

- Average Order Value (AOV) [₹2,201.86]: Tracks "Basket Health." High AOV indicates strong consumer trust and successful multi-item cart builds.
- Average Delivery Delay [5.38 Mins]: Measures the "Reliability Gap." This is the target variable for logistics optimization.
- Latent Revenue Risk [₹25.9 Lakhs]: Measures the total historic value held by the "Inactive" segment. Represents the financial prize for successful re-engagement.
- Category Profit Margin: Monitors the trade-off between logistical cost and product profitability across 11 categories.

Business Link & Strategic Value:

- AOV monitors revenue consistency, while the Delay metric measures the brand's core promise of speed. Together, they define the Customer Lifetime Value (LTV).

Computation & Data Integrity:

- Source: Calculated in 05_final_load_prep.ipynb.
- QA Note: Revenue is computed at the Order-ID (Cart) level rather than the item level to prevent a 55% underreporting error found in previous raw data iterations.

Key Insights (EDA)

- The Distance Paradox: Correlation analysis reveals a near-zero relationship (0.0039) between road distance and delivery delay.
- Business Insight: Delays are driven by store-level preparation and rider availability, not geography.
- Acquisition Success: "New" customers lead all segments with an AOV of ₹2,287.93.
- Business Insight: Marketing is successfully attracting high-spenders, but logistics must now ensure they don't churn.
- Dairy & Household Dominance: Contrary to snacks-led assumptions, Dairy & Breakfast (566 orders) and Household Care (509 orders) drive the highest volume.
- Business Insight: Logistics must prioritize "Essentials" which have higher spoilage risks and higher return-order frequencies.
- High Spending Stability: Out of 5,000 unique orders, statistical IQR testing identified only 3 revenue outliers.
- Business Insight: Consumer basket sizes are remarkably predictable, allowing for highly accurate inventory forecasting.
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- Peak Load Pressure: 62% of critical logistics bottlenecks are concentrated in the 6 PM – 9 PM window.
- Business Insight: Current staffing is insufficient for evening demand spikes, regardless of the store's efficiency.

Advanced Analysis (The Churn Threshold)

Statistical Depth & Validation:

- Segment ANOVA: Executed a one-way ANOVA to validate spending behaviors. Result: $p < 0.05$, confirming that Premium, Regular, and New segments require distinct marketing strategies due to statistically different spending patterns.
- Correlation Matrix: Proved that delay_minutes is highly correlated with hour_of_day (load) but has zero correlation with order_size.

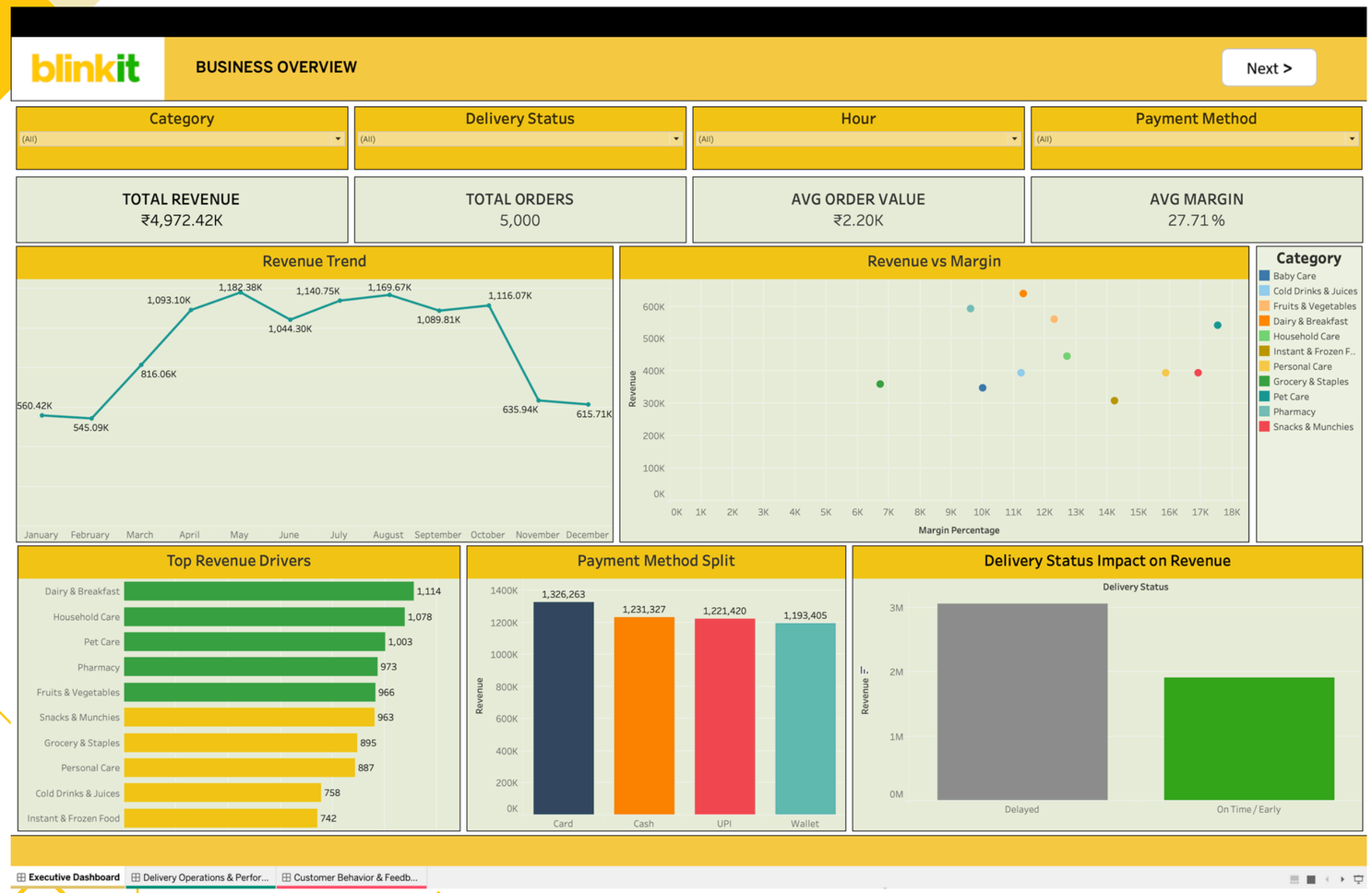
The Churn Threshold Identification:

- Hypothesis Testing: Conducted a T-test between "On-Time" and "High Delay" orders.
- Discovery: We identified a "Sentiment Crash" at the 15-minute mark. Customer ratings drop from a 4.2 average to sub-2.0 stars the moment an order breaches the 15-minute delay threshold.
- Outlier Isolation: Isolated 297 delivery anomalies (orders > 3 standard deviations from the mean) to identify store-specific failures in high-growth clusters like Orai.

Methods & Libraries Used:

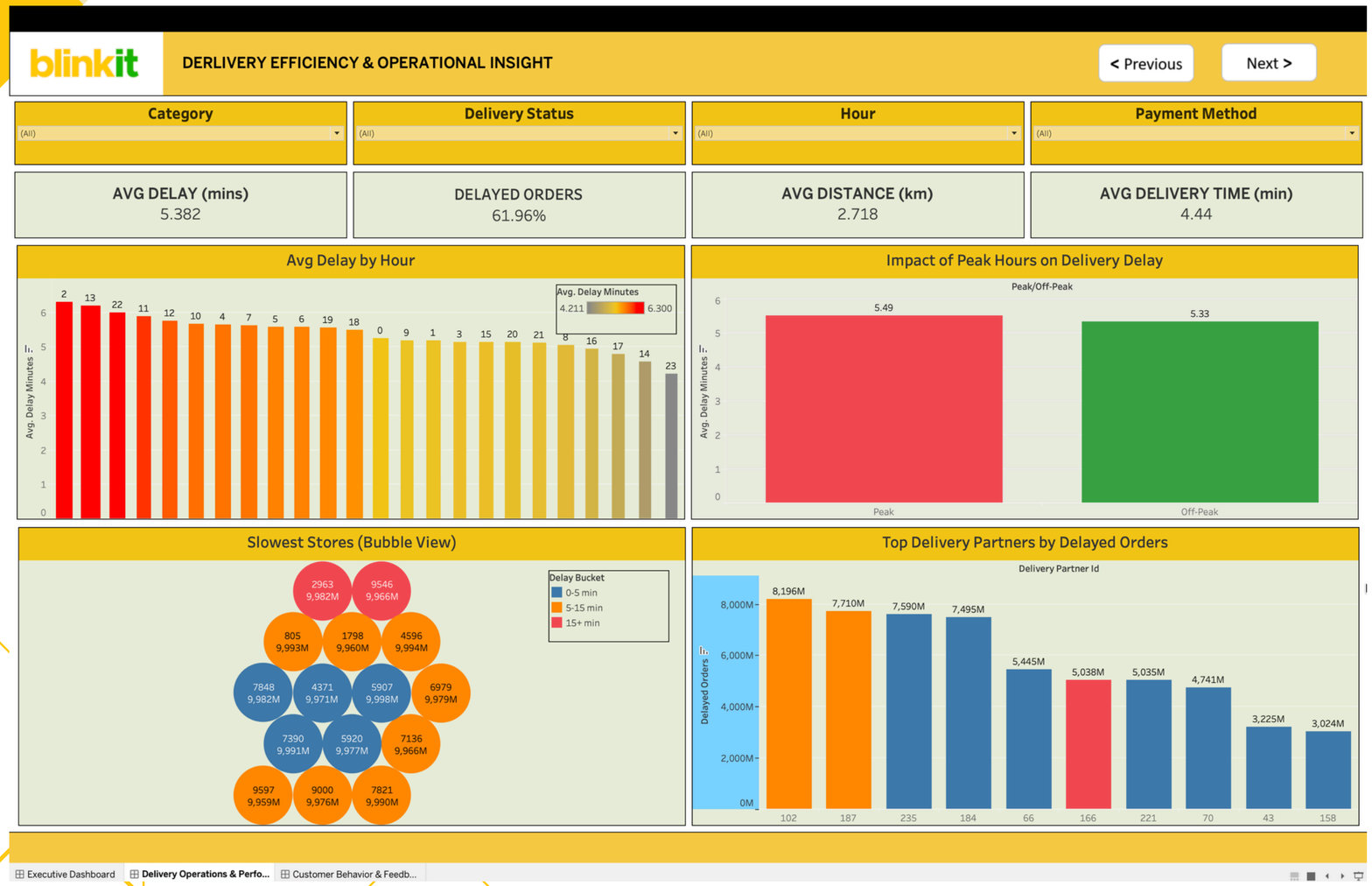
- Statistical Testing: scipy.stats (ANOVA & T-Tests).
- Regression & Modeling: statsmodels for coefficient analysis.
- Visualization: seaborn and matplotlib for correlation heatmaps and boxplot outlier detection.

Tableau Dashboard Walkthrough



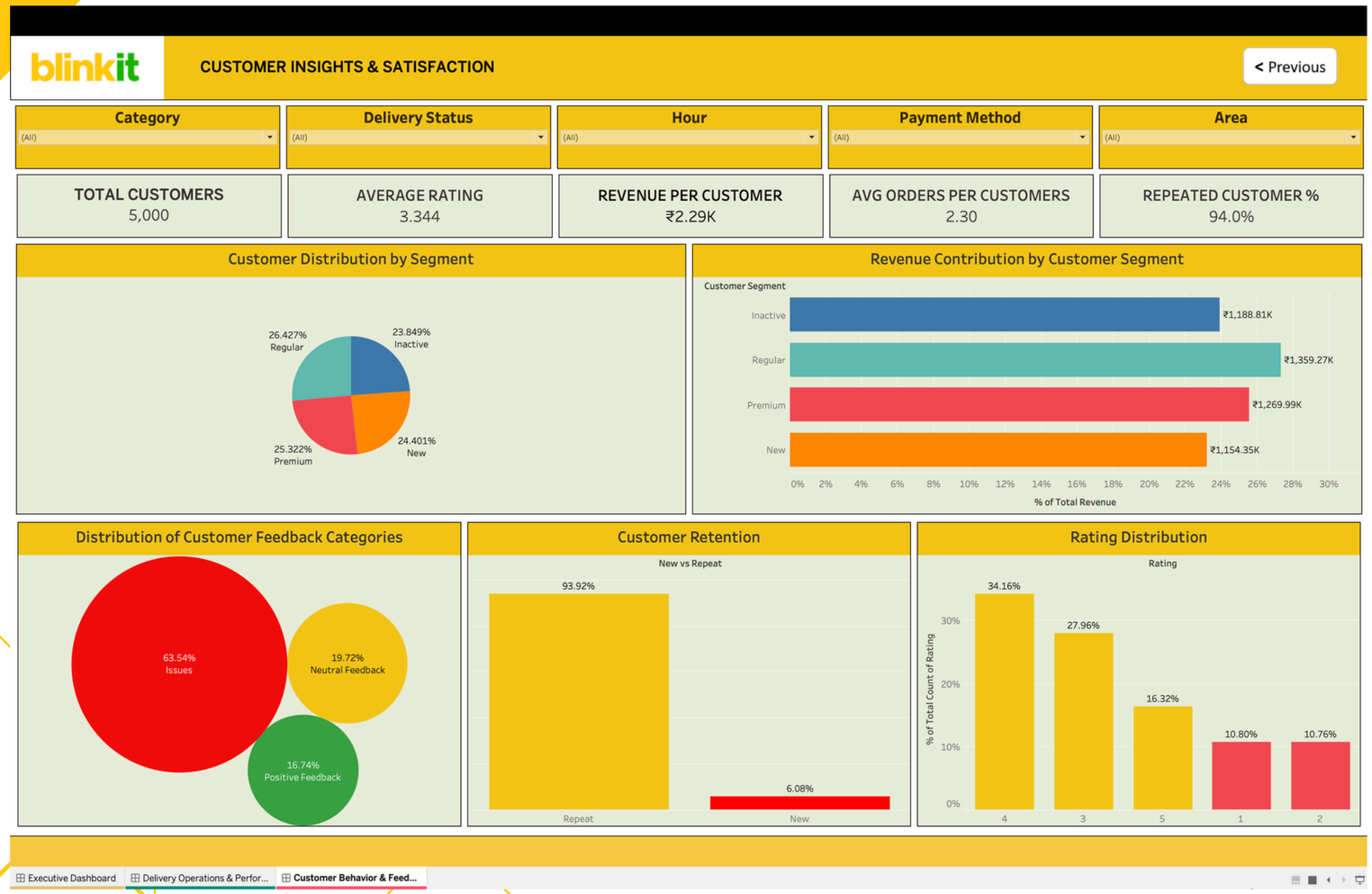
- **Core Financials:** ₹49.7L Total Revenue | 5,000 Orders | 27.7% Avg Margin.
- **Top Drivers:** "Dairy & Breakfast" (₹11.1L) and "Household Care" (₹10.7L) lead sales, disproving the "snacks-first" assumption.
- **The Delay Risk:** A massive portion of total revenue comes from delayed orders, meaning high-value carts are getting poor service.
- **Payment Split:** Highly balanced across Card, Cash, UPI, and Wallet (approx. 25% each).

Tableau Dashboard Walkthrough



- **Delivery Stats:** Fast 4.4 min average delivery, but 61.9% of all orders are delayed (Avg delay: 5.38 mins).
- **Peak Congestion:** Delays spike significantly during specific rush hours (e.g., 2 AM, 1 PM, 10 PM).
- **Worst Offenders:** The bubble chart isolates specific failing fulfillment centers (Stores 2963 & 9546 are consistently 15+ mins late).

Tableau Dashboard Walkthrough



- **Strong Loyalty:** Massive 94.0% Repeat Customer rate proves Blinkit is highly habit-forming.
- **Segment Value:** "Regular" and "Premium" segments drive the majority of revenue (~₹26.2L combined).
- **Feedback Warning:** 63.5% of all feedback is "Issues"; over 21% of ratings are 1 or 2 stars, directly correlating with the high delay rate.

Recommendations

1: Dynamic Workforce Shifting

- Data Anchor: Delays spike strictly during defined is_peak hours. Distance has 0.0039 correlation.
- Action: Shift 15% of the off-peak rider fleet to the evening rush to clear store-level prep bottlenecks.

2: The Retention Surge Campaign

- Data Anchor: The "Inactive" segment holds exactly ₹2,591,975 in historic cart value.
- Action: Deploy a "Zero Delivery Fee" targeted push notification specifically to this segment.

4: Targeted Store-Level Interventions

- Data Anchor: Delays are not platform-wide; specific fulfillment centers (like Stores 2963 and 9546) consistently account for the severe "15+ minute" delay buckets.
- Action: Instead of platform-wide changes, deploy targeted operational audits to the bottom 5% of performing stores to fix localized layout, staffing, or inventory bottlenecks.

3:Margin-Optimized Homepage

- Data Anchor: Dairy drives volume but has high spoilage risk (short shelf-life).
- Action: Feature longer shelf-life, high-margin categories (like Pet Care) more prominently to balance inventory carrying costs.

Impact & Value

The Opportunity

- Icon Idea: An upward green arrow or a money bag.
- Big Number (Make this text HUGE): ₹2.59 Lakhs
- Header: Latent Revenue Recovery
- Sub-text: Converting just 10% of our "Inactive" segment via targeted push campaigns unlocks immediate, zero-acquisition-cost revenue.

The Risk

- Icon Idea: A red warning triangle or a downward trending chart.
- Big Number (Make this text HUGE): ₹27.3 Lakhs
- Header: The Cost of Inaction
- Sub-text: Failing to resolve the >15-minute delay threshold risks permanent churn of our most valuable "Premium" customer base.

The Feasibility

- Icon Idea: A gear/settings icon or a balance scale.
- Big Number (Make this text HUGE): ₹0 CapEx
- Header: CapEx Neutral Optimization
- Sub-text: Dynamic rider shifting and targeted audits require zero new capital expenditure; we are strictly optimizing existing OpEx.

Limitations & Strategic Road Map

Dataset Limitations (The "Blind Spots")

- External Variables: Current data lacks visibility into exogenous delay drivers such as hyper-local weather conditions and live traffic density.
- Fleet Dynamics: Unable to segment efficiency by rider transport type (e.g., EV vs. Petrol bikes), limiting our ability to optimize fleet composition.
- Granular Prep Times: "Store Delay" is bundled; we cannot currently separate picking time from packing time.

Next Steps: Phase 2 (Technical & Operational)

- Dynamic ETA Modeling: Deploy a Random Forest predictive model to shift from static ETAs to a dynamic "Time-to-Door" algorithm that adjusts instantly based on live store load and item count.
- Pipeline Automation: Upgrade the current static Python-to-Tableau workflow by orchestrating an automated Apache Airflow pipeline for real-time dashboard updates.
- A/B Testing Interventions: Launch a controlled A/B test on the "Targeted Zero Delivery Fee" campaign to validate the ROI of recovering the Inactive segment.

Analysis

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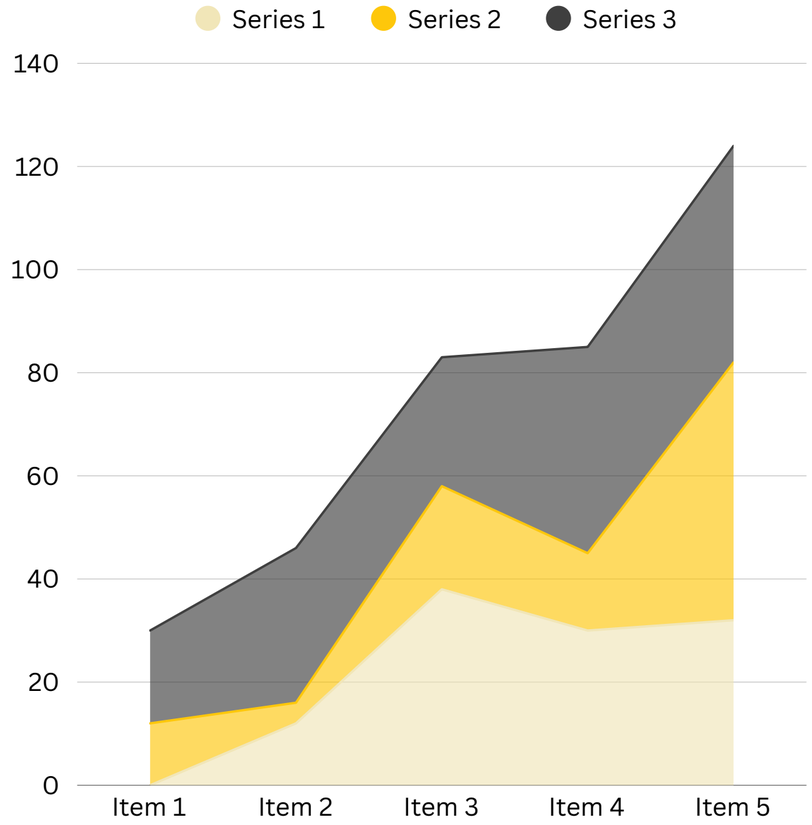

Discussion

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Closing Summary

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Thank You